

DATA ANALYSIS HACKATHON

Formula 1 World Championship Analysis

1950 – 2022 | Power BI | 4 Hours

The Scenario

You just joined the analytics team of a top Formula 1 constructor. The team's performance director needs a full intelligence report before the next season kicks off. You have 4 hours, 4 data files, and Power BI.

Your mission: build a dashboard that answers the questions the director actually cares about — not just charts, but decisions.

Section 1 — The Dataset

You will receive **4 CSV files** covering Formula 1 races from 1950 to 2022. Each file represents one table in the data model you must build.

File	Rows	Key Columns
fact_Results.csv	25,840	season_year, round_number, driver_name, team, position, points, notes, laps
dim_Races.csv	1,079	season_year, round_number, grand_prix, race_date, circuit_name, country
dim_Drivers.csv	855	driver_name, nationality, date_of_birth, race_number
dim_Circuits.csv	76	circuit_name, location, country, lat, lng, altitude

Known Data Quality Issues

These issues are intentional. Part of your score depends on how you handle them in Power Query.

- **\N** in multiple columns — represents missing data, must be replaced with null
- position column — contains 'R', 'F', and numbers mixed together
- points column — stored as decimal (9.0) instead of integer
- race_date — stored as text, must be converted to Date type
- driver_name — some names have extra spaces (use TRIM)
- Q1/Q2/Q3 columns — empty before 2003 (historical, not an error)

Section 2 — Data Modeling

Your first task is to build the **Snowflake Schema** below. Do NOT work from a single flat table.

```
fact_Results  →  dim_Races  (on: season_year + round_number — composite key)
fact_Results  →  dim_Drivers  (on: driver_name)
dim_Races  →  dim_Circuits  (on: circuit_name)
```

⚠ The dim_Circuits table connects to dim_Races — not to fact_Results directly. This is a Snowflake Schema, not a pure Star Schema. Your DAX must navigate this correctly.

Section 3 — Required DAX Measures

Mandatory KPIs (7 pts)

All 3 measures are required. Name them exactly as shown.

- **KPI 1 — Points Per Race (2 pts)**

  Logic: Total points divided by the count of races (rows) in the `fact_Results` table.

- **KPI 2 — Win Rate % (2 pts)**

  Logic: The percentage of races where the driver finished in 1st position relative to their total race entries.

- **KPI 3 — Podium Rate % (3 pts)**

```
Podium Rate % =   DIVIDE(     COUNTROWS(       FILTER(fact_Results,
fact_Results[position] IN {"1", "2", "3"})     ),
COUNTROWS(FILTER(fact_Results,       NOT(fact_Results[position] IN
{"R","F"})))   ) * 100
```

Bonus Measure — choose ONE (3 pts)

- **Option A — Championship Gap (3 pts)**

  Logic: Calculate the point difference between a specific driver and the current leader (rank 1) of that season.

- **Option B — Constructor Dominance % (3 pts)**

  Logic: The share of total season points earned by a single team (constructor) compared to the total points available in that season.

Section 4 — Dashboard Requirements

Required Pages

- Overview Page — high-level KPIs, season filter, top 5 drivers by total points
- Driver Analysis — per-driver deep dive, Win Rate %, Podium Rate %
- Circuit & Geography — map visual using lat/lng from dim_Circuits
- Team Performance — constructor comparison across seasons

Mandatory Creative Feature — pick ONE

> **Option 1 — Slicer Panel:** A collapsible pop-out filter panel using Bookmarks
> **Option 2 — Insight Toggle:** Switch between Bar Chart and Line Chart views using Bookmarks
> **Option 3 — What-If Parameter:** 'What if fastest lap bonus points increased to X?'
> **Option 4 — Drill-Through:** Right-click a driver → see full race-by-race breakdown

Section 5 — Business Insights

Your dashboard must include a **dedicated Insights page** or **Smart Narrative visuals** that answer the following questions as if presenting to the team director:

- Which driver shows the highest consistency across multiple seasons?
- Which circuit produces the most DNF retirements? What could explain this?
- Is there a team that dominates specific circuit types (street vs. track)?
- What trend do you see in championship competitiveness over the decades?
- If you had to recommend ONE driver for a future contract based purely on data — who and why?

Insights must be written in plain English (or Arabic). Data without context is not an insight.

Section 6 — Grading Breakdown

Element	Score	Notes
Data Cleaning & ETL (Power Query)	**5 pts**	TRIM, Replace \N, fix Position column, data types
Data Modeling (Relationships)	**5 pts**	Snowflake Schema — 3 relationships correct
DAX Measures (KPIs)	**7 pts**	3 mandatory KPIs implemented correctly
DAX Bonus Measure	**3 pts**	Championship Gap or Constructor Dominance %
UI/UX & Visualization	**5 pts**	Clean dashboard, consistent theme, readable visuals
Business Insights & Storytelling	**5 pts**	Actionable insights, not just charts

Element	Score	Notes
Creative Feature (Bonus)	**3 pts**	Bookmarks / What-If / Drill-Through / Tooltip
Total	**33 pts**	30 pts mandatory + 3 pts bonus

Section 7 — Recommended Timeline

Total duration: 4 hours. Stick to the plan — the biggest mistake is spending 2 hours on cleaning and having no time for insights.

Phase	Time	Task
01	**60 min**	ETL & Data Cleaning — Power Query: Replace \N, fix Position types, TRIM driver names, fix data types
02	**30 min**	Data Modeling — Build the 3 Relationships in the Snowflake Schema
03	**60 min**	DAX Measures — Build the 3 mandatory KPIs + 1 optional bonus
04	**60 min**	Dashboard & Visualization — Build pages, apply theme, add 1 creative feature
05	**30 min**	Insights & Storytelling — Write your business recommendations

Section 8 — Submission Requirements

- One Power BI file (.pbix) — containing all 4 data sources embedded
- File naming: FirstName_LastName_F1_Hackathon.pbix
- Ensure the .pbix opens without any data refresh errors
- All relationships must be visible in Model View
- No external connections — data must be embedded

Winner Selection Criteria

Highest total score • Data accuracy & modeling correctness

Dashboard clarity & visual storytelling • Quality of business insights

Duration: 4 hours | Power BI Desktop required | Good luck!